APPENDIX

HARSHAW MOUNTED SCINTILLATION PHOSPHORS

Harshaw STANDARD LINE detector assemblies consist of right circular cylindrical crystals that have been machined, surfaced, and mounted in precision housings. These assemblies are readily coupled to a wide variety of photomultiplier tube types and offer distinct advantages in versatility and economy.

Each STANDARD LINE crystal housing is normally fabricated from aluminum or low background stainless steel and is provided with a hermetically sealed optical window of quartz, Vycor, Pyrex or glass. Crystal housings can also be fabricated from a variety of other alloy materials including OFHC copper and cold-rolled steel upon special request.

Type D assemblies are the basic STANDARD LINE detector. Temporary mounting of these units to photomultiplier tubes can be accomplished using optical coupling compounds and black tape as a light seal. More permanent mounting techniques incorporate epoxied flanges that couple directly to bolt-down magnetic shields. **Type A** assemblies resemble D style detectors but are fabricated with spun flanges. This type construction permits the use of slip-on adapter rings that couple to bolt-down shields, making the entire assembly fully demountable.

Type AW Top Well detector assemblies permit high counting efficiencies with small volume samples. These units are also fabricated with spun flanges for use with bolt-down shields. **Type F** well assemblies are manufactured with "lip-and-groove" flanges to facilitate the addition of a rubber "O-ring" between the crystal housing and the shield mounting hardware. The ring protects the spun aluminum housing from damage that may result from over-tightening the shield mounting hardware. These detectors utilize a unique .004" sheet reflector around the well to minimize crystal-to-housing dimensions and gamma ray absorption.

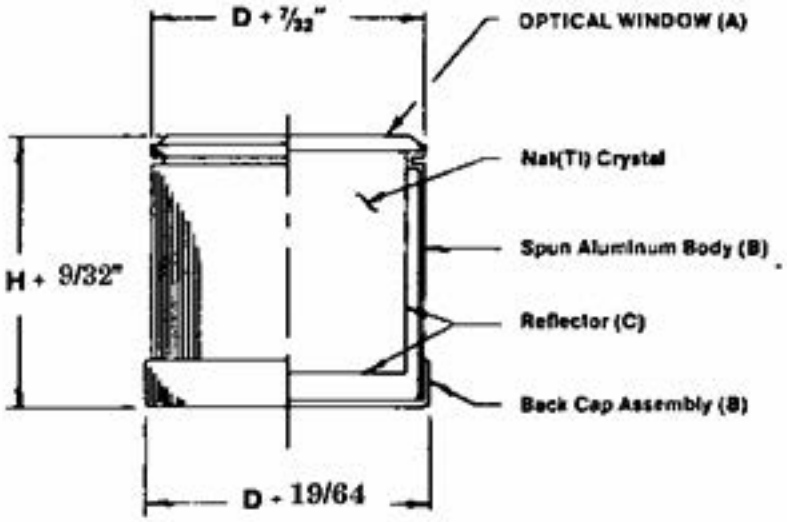
Type AF (through-side-hole) detectors are right circular cylindrical crystals with holes machined perpendicular to the crystal axis. These units are widely used for small volume samples and in flow studies.

Type H assemblies are offered with beryllium or aluminum entrance windows for low energy applications. The crystals used in these detectors are either thinly cleaved or machined and polished, depending upon application requirements. The cleaved crystals are preferred for applications in the soft X-ray region below 10keV since the X-rays are absorbed in the surface layer of the crystal.

Type HG assemblies resemble H style detectors but are fabricated with one-piece spun aluminum housings. These units can be adapted to a wide range of crystal sizes up to three inches. Sheet reflector material is used exclusively in these assemblies to minimize the dimensions between the crystal and the housing and reduce gamma ray absorption.

Type PF and PA assemblies are ruggedized for applications where mechanical shock, large thermal gradients or wide temperature extremes may be encountered. Harshaw guarantees Type PF detectors to withstand temperatures between -50°C and +150°C and 50g's shock. These assemblies are widely used in well logging applications and feature one-piece stainless steel housings. PA assemblies are fabricated with one-piece spun aluminum housings and are widely used as medical probes and in environmental applications. Type PA detectors are guaranteed to operate per specifications in a temperature range of -21°C to +65°C.

Each STANDARD LINE detector assembly carries a two-year warranty against detector failure due to faulty construction, materials or hermetic seals used in the crystal assembly.



TYPE(D) – CRYSTAL

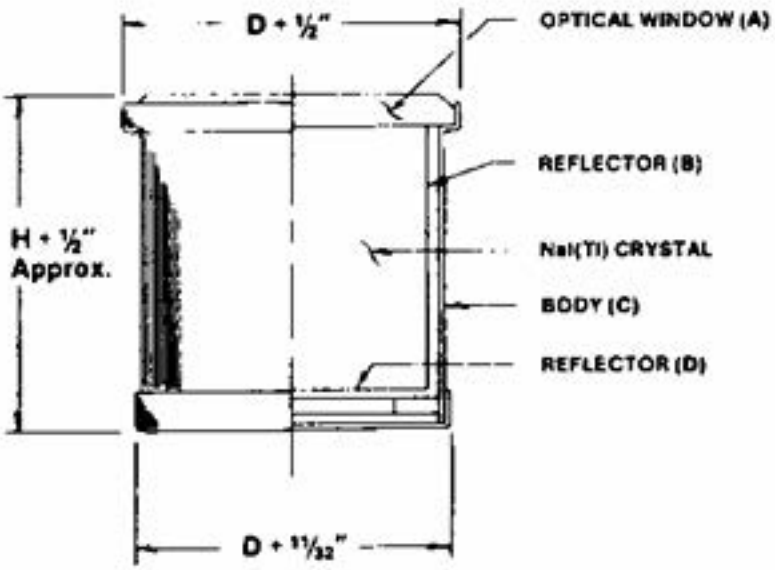
Intended Application:

General laboratory research, medical and industrial research

Crystal Size Range:
DIAMETER — ½" to 5" in increments of ¼"
HEIGHT — ¼" to 5" in increments of ¼"

A ⅛" glass — quartz available on special order
B .032" Type 1100 aluminum
C Packed Al_2O_3 approximately 1/16" thick

D — Crystal Diameter
H — Crystal Height

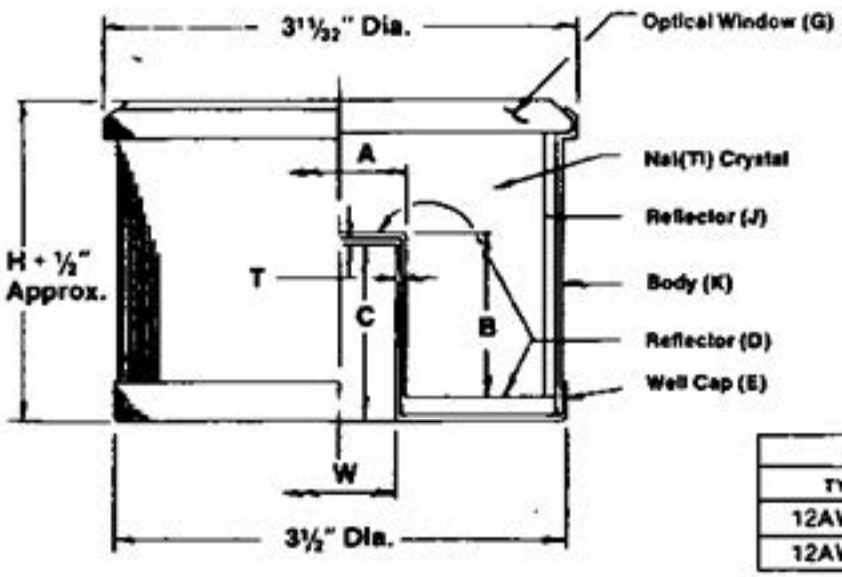


Intended Application:

High gamma energy or total spectrometry — also recommended for low level counting

A ⅛" glass (quartz available upon request)
B Packed Al_2O_3 approximately 1/16" thick
C .032" Type 1100 aluminum
D HR-15 sheet reflector

D — CRYSTAL DIAMETER (up to 5")
H — CRYSTAL HEIGHT



H — Crystal Height

Intended Application:

High gamma counting efficiency for small volume solid or liquid samples

G ⅛" glass — quartz available upon request
J Packed Al_2O_3 approximately 1/16" thick
K .032" Type 1100 aluminum
D HR-15 sheet reflector
E .032" aluminum

STANDARD WELL SIZES:*

	XTAL DIMENSIONS		WELL DIMENSIONS		
TYPE NO.	A = DIA.	B = DEPTH	C = DEPTH	W = DIA.	T = THICKNESS
12AW( )-W3	.781	2.000	2.250	.656	.032
12AW( )-W4	1.125	2.000	2.250	1.000	.032

*Also available in additional sizes up to 5" dia. with special well dimensions upon request.

Material	Wavelength of Maximum Emission (nm)	Decay Constant (μ s)*	Scintillation Cutoff Wavelength (nm)	Index of Refraction**	Density (g/cc)	Hygroscopic	γ Scintillation Conversion Efficiency (%)***
NaI(Tl)	410	.23	320	1.85	3.67	Yes	100
CaF_2(Eu)	435	.9	405	1.47	3.19	No	50
CsI(Na)	420	.63	300	1.84	4.51	Yes	85
CsI(Tl)	565	1.0	330	1.80	4.51	No	45
^{6}LiI(Eu)	470[1]	.94	450	1.96	3.49	Yes	35
Plastics[2]	350-450	.002-.020[3]	Varies	Varies	1.06[3]	No	5[3]
Liquids[2]	350-450	.002-.008[3]	Varies	Varies	0.86[3]	No	5[3]

TABLE I

* Room temperature, best single exponential decay constant, $I_0 e^{-\lambda t}$

** At emission maximum

*** Referred to NaI(Tl) with S-11 photocathode response

1 Primarily used for neutron detection

2 Presented for comparison only, Harshaw does not manufacture

3 Dependent upon composition, values given are typical

FIGURE 1

Emission spectra of NaI(Tl), CsI(Tl) and CsI(Na) compared to the relative spectral sensitivities of S-11 and bialkali photomultiplier tube types. The emission curves have been normalized to 100% for illustrative purposes. Harshaw Research Laboratory Report.

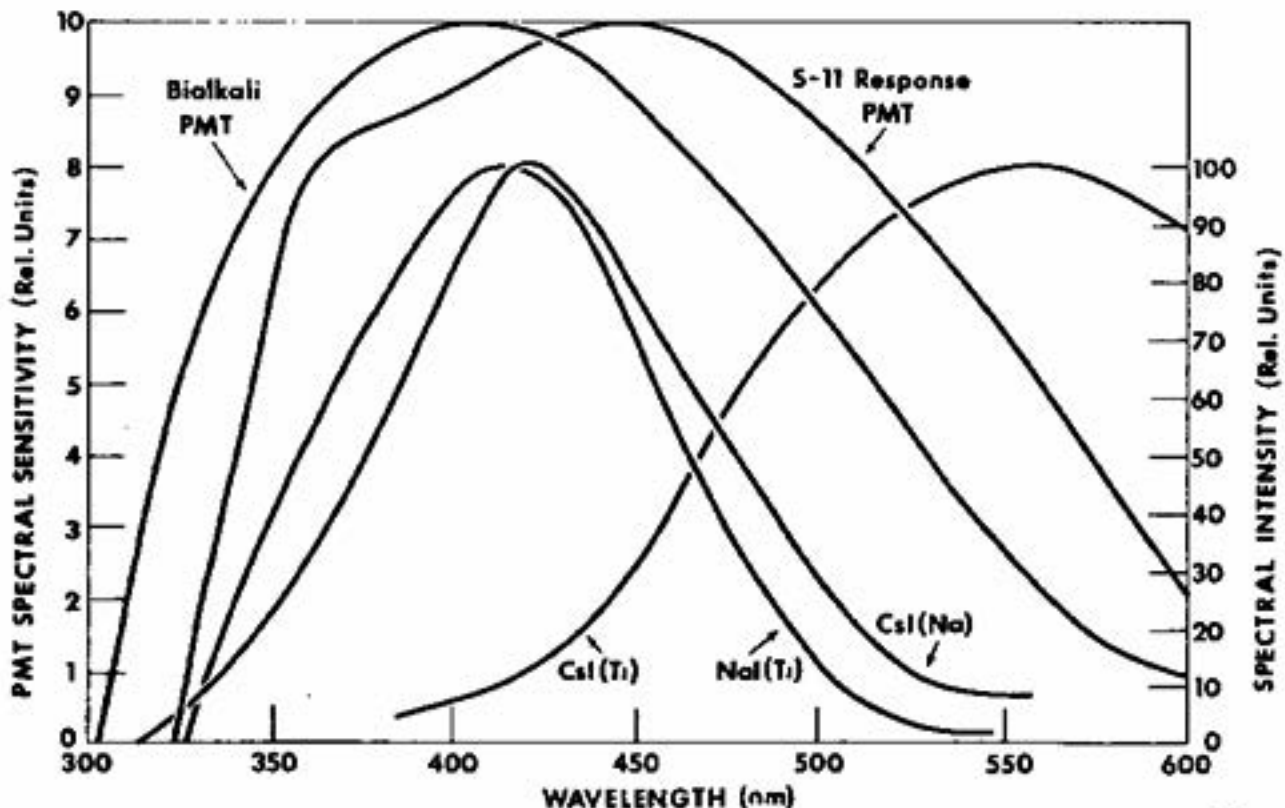


FIGURE 3

Efficiency is defined as ($E = N/N_0 \times 100\%$) where N_0 is the total number of gamma rays per second radiated by the source and N represents the number of interactions within the detector crystal. From "Calculated Efficiencies of NaI Crystals" by E.A. Wolicki, R. Jastrow and F. Brooks, NRL Report 4833.

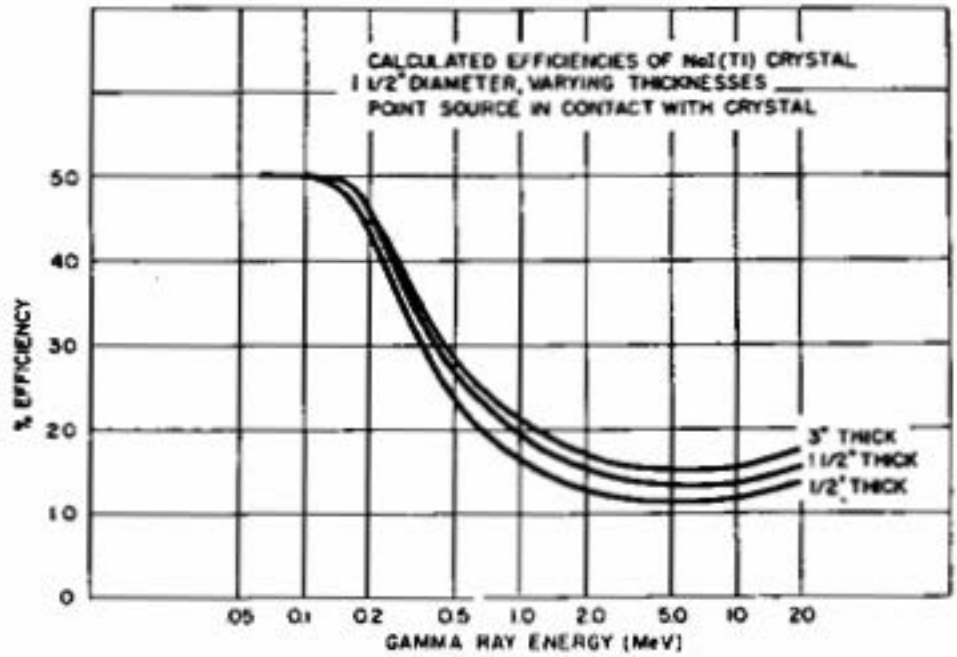
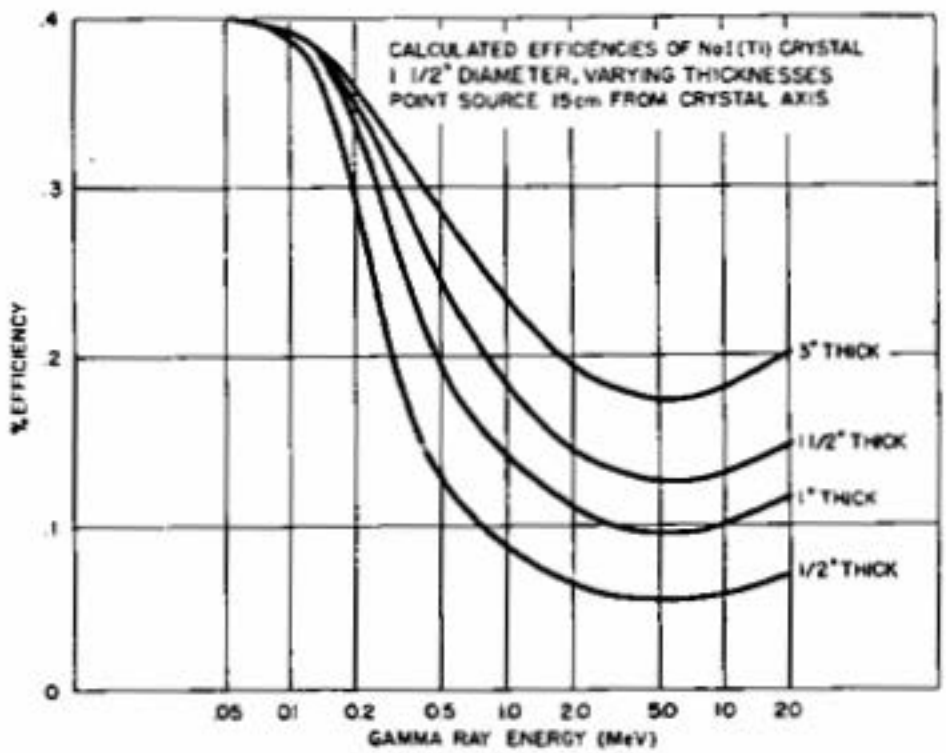


FIGURE 4

From "Calculated Efficiencies of NaI Crystals" by E.A. Wolicki, R. Jastrow and F. Brooks, NRL Report 4833.



These efficiencies (E) are given by ($E = N/N_0 \times 100\%$) where N_0 is the total number of gamma rays per second radiated by the source and N represents the number of interactions within the crystal. Actual efficiency depends upon absorption by the detector housing and the fraction of interactions that are counted by the electronic circuitry (the electronics in this case are assumed to have 100% counting efficiency). The effect of absorption by the detector housing is important at energies below 100keV and is illustrated in Figure 5 for several housing materials.

Figure 3 illustrates the effect of varying crystal thickness when the source is in contact with the crystal. Note that efficiency is not a strong function of crystal thickness for this geometry. In contrast, Figure 4 illustrates a rapid change of efficiency with crystal thickness when the source is 15cm distant from the crystal face. The passage of gamma rays in this instance is nearly perpendicular to the crystal and the increase is roughly exponential with thickness.

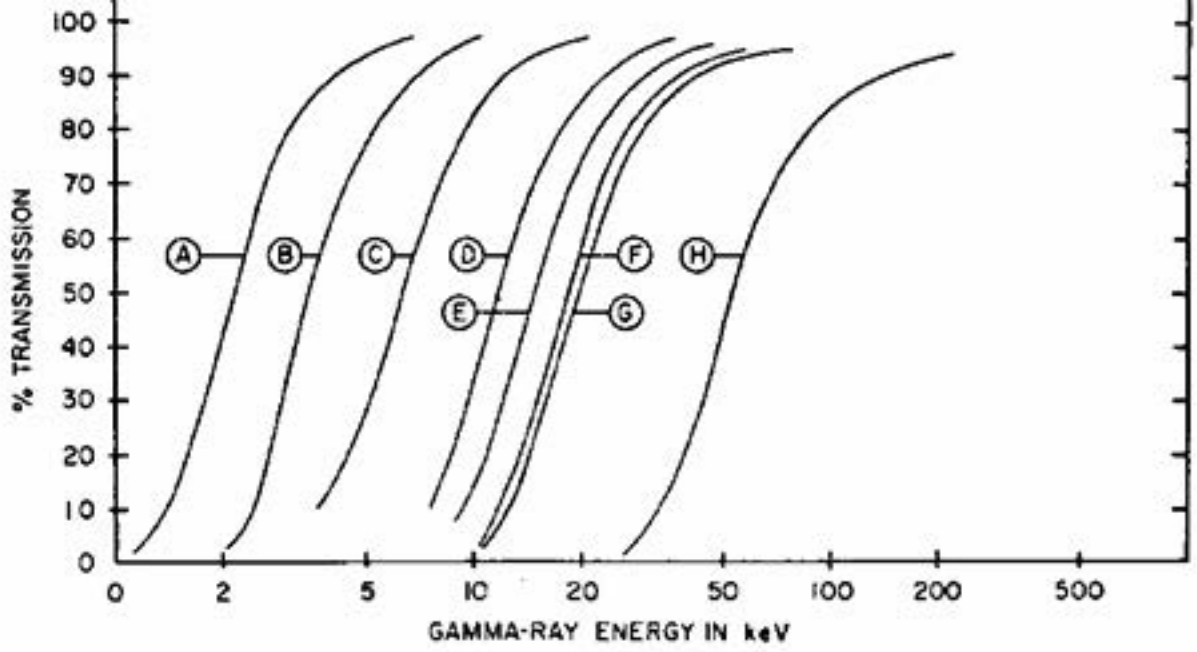


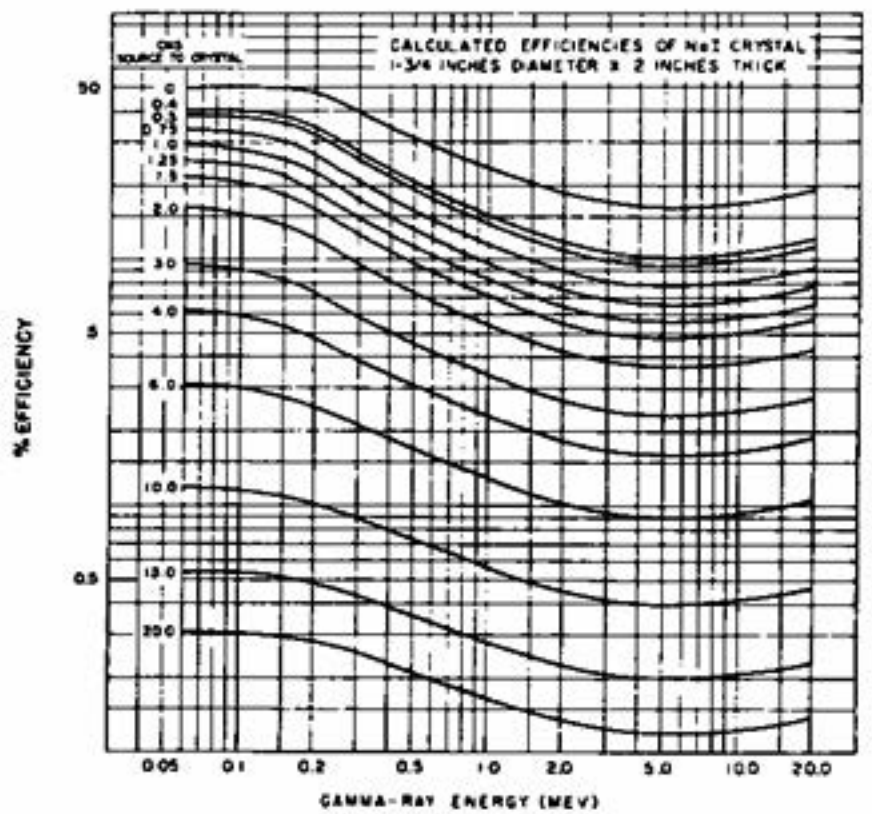
FIGURE 5

Per cent transmission vs. gamma ray energy for various detector housing and entrance window materials. Harshaw Research Laboratory.

A .002" Be
B .010" Be
C .001" Al
D .005" Al
E .010" Al
F .015" Al
G .019" Al
H .019" STAINLESS STEEL

FIGURE 6

Calculated Efficiencies of a NaI(Tl) crystal 1¾" diameter and 2 inches thick. From "Calculated Efficiencies of NaI Crystals" by E.A. Wolicki, R. Jastrow and F. Brooks, NRL Report 4833.



In many applications it is desirable to discriminate against background radiation or other spurious events. In these instances, it is necessary to count only full energy or photopeak counts. Photopeak efficiency can be obtained from Figures 3, 4, 6 and 7 by multiplying the values given by the appropriate photofraction (peak-to-total ratio). Crystal diameter, length and the direction of the incident radiation determine the photofraction at a given energy. Figures 11, 12 and 13 give typical photofractions for crystals of various sizes as a function of energy and source geometry.

An example of a photopeak counting efficiency calculation:

Detector ..... NaI(Tl) 3" x 3"
Energy of Source 662keV
Source Distance ..... 10cm from detector face

From Figure 7 the total per cent efficiency is 2%. The photofraction from Figure 12 is 0.60. Thus, E (photopeak) = (2%) x (0.60) = 1.2%. The calculation shows approximately 1% of the counts emitted by the source at a distance of 10cm will be counted in the energy region near the photopeak.

The relative light output of all scintillators is a function of temperature. The inorganic crystals and the alkali halides in particular exhibit similar response to temperature as illustrated in Figure 14. This effect is significant in a number of applications such as oil well logging, space exploration and environmental surveillance.

In 1972 Harshaw introduced a unique polycrystalline scintillation material called POLYSCIN™ NaI(Tl). The development of this material was the result of new extrusion and forging techniques permitting detector sizes and geometries previously unobtainable with single crystal ingots.*

In contrast to single crystals, POLYSCIN™ NaI(Tl) is composed of many small crystallites ranging in size from less than 1mm to several millimeters. Because these crystallites are in close contact throughout the material, POLYSCIN™ crystals are fully equivalent to single crystals in optical and scintillation performance. Any light scattering at the crystallite boundaries is negligible. The density of POLYSCIN™ crystals is identical to single NaI(Tl) crystals.

FIGURE 14

Temperature response of NaI(Tl), CsI(Na) and CsI(Tl). Relative light output has been normalized to 100% for illustrative purposes. Harshaw Research Laboratory Report.

